

```

1 class SGTree {
2     public:
3         vector<int> seg;
4         SGTree(int n) {
5             seg.resize(4 * n + 1);
6         }
7         void build(int idx, int left, int right, vector<int> &v) {
8             if (left == right) {
9                 seg[idx] = v[left];
10                return;
11            }
12            int mid = (left + right) / 2;
13            build(2 * idx + 1, left, mid, v);
14            build(2 * idx + 2, mid + 1, right, v);
15            seg[idx] = seg[2 * idx + 1] + seg[2 * idx + 2];
16        }
17        void update(int idx, int left, int right, int ind, int val, ) {
18            if (left == right) {
19                seg[idx] = val;
20                return;
21            }
22            int mid = (left + right) / 2;
23            if (ind <= mid)
24                update(2 * idx + 1, left, mid, ind, val);
25            else
26                update(2 * idx + 2, mid + 1, right, ind, val);
27            seg[idx] = seg[2 * idx + 1] + seg[2 * idx + 2];
28        }
29        int query(int idx, int left, int right, int l, int r) {
30            if (r < left || l > right)
31                return 0;
32            else if (left >= l && r >= right) {
33                return seg[idx];
34            }
35            int mid = (left + right) / 2;
36            int a = query(2 * idx + 1, left, mid, l, r);
37            int b = query(2 * idx + 2, mid + 1, right, l, r);
38            return a + b;
39        }
40    };

```